

Econometrics Home Task 4

Subject code: DA242

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1 Questions

1. Five observations on x and y are as follows:

$$x = [4, 1, 5, 8, 2], \quad y = [6, 3, 12, 15, 4]$$

Assume a linear model with heteroscedasticity of the form:

$$\sigma_t^2 = \sigma^2 x_t^2$$

Calculate the Generalized Least Squares (GLS) estimates of the regression coefficients.

2. Estimate a linear relation by applying GLS to the following data:

$$x = [2, 3, 1, 5, 9], \quad y = [4, 7, 3, 9, 17]$$

The variance matrix for the disturbances is:

$$\text{Var}(e) = \sigma^2 \text{diag}\{0.10, 0.05, 0.20, 0.30, 0.15\}$$

3. Consider the following regression-through-the-origin model:

$$y_i = \beta x_i + e_i, \quad i = 1, 2$$

where

$$e_1 \sim N(0, \sigma^2), \quad e_2 \sim N(0, 2\sigma^2), \quad e_1 \perp e_2$$

and

$$x_1 = +1, \quad x_2 = -1$$

Obtain the Weighted Least Squares (WLS) estimate of β and its variance.

2 Solutions

2.1 Solution Q1.

We are given the data:

$$x = [4, 1, 5, 8, 2], \quad y = [6, 3, 12, 15, 4]$$

and the linear regression model with heteroscedasticity:

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad u_t \sim N(0, \sigma^2 x_t^2)$$

Step 1: Transform the model

Divide both sides by x_t to stabilize the variance:

$$\frac{y_t}{x_t} = \beta_0 \frac{1}{x_t} + \beta_1 + \frac{u_t}{x_t}, \quad \frac{u_t}{x_t} \sim N(0, \sigma^2)$$

Define:

$$Y_t^* = \frac{y_t}{x_t}, \quad X_t^* = \frac{1}{x_t}, \quad \epsilon_t = \frac{u_t}{x_t}$$

Then the transformed regression is:

$$Y_t^* = \beta_0 X_t^* + \beta_1 + \epsilon_t$$

Step 2: Compute transformed variables

$$Y^* = \frac{y}{x} = [1.5, 3.0, 2.4, 1.875, 2.0], \quad X^* = \frac{1}{x} = [0.25, 1.0, 0.2, 0.125, 0.5]$$

Step 3: Compute means

$$\bar{X}^* = \frac{0.25 + 1.0 + 0.2 + 0.125 + 0.5}{5} = 0.415$$

$$\bar{Y}^* = \frac{1.5 + 3.0 + 2.4 + 1.875 + 2.0}{5} = 2.155$$

Step 4: GLS estimates

The slope:

$$\hat{\beta}_0 = \frac{\sum (X_t^* - \bar{X}^*)(Y_t^* - \bar{Y}^*)}{\sum (X_t^* - \bar{X}^*)^2} = \frac{0.61775}{0.507} \approx 1.219$$

The intercept:

$$\hat{\beta}_1 = \bar{Y}^* - \hat{\beta}_0 \bar{X}^* = 2.155 - 1.219 \cdot 0.415 \approx 1.649$$

Step 5: GLS regression equation

$$y_t \approx 1.649 + 1.219x_t$$

2.2 Solution Q2.**Step 1: GLS formula**

For a linear regression model $y = X\beta + e$ with $\text{Var}(e) = \Sigma$, the GLS estimator is:

$$\hat{\beta}_{GLS} = (X^\top \Sigma^{-1} X)^{-1} X^\top \Sigma^{-1} y$$

Here,

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 1 \\ 1 & 5 \\ 1 & 9 \end{bmatrix}, \quad y = \begin{bmatrix} 4 \\ 7 \\ 3 \\ 9 \\ 17 \end{bmatrix}, \quad \Sigma = \sigma^2 \text{diag}\{0.10, 0.05, 0.20, 0.30, 0.15\}$$

Step 2: Compute weights

Since σ^2 cancels, define weights:

$$w_i = \frac{1}{\text{Var}(e_i)} = [10, 20, 5, 3.333, 6.667]$$

Take square roots for weighted regression:

$$\sqrt{w_i} \approx [3.1623, 4.4721, 2.2361, 1.8257, 2.58199]$$

Step 3: Weighted variables

$$X^* = W^{1/2} X = \begin{bmatrix} 3.1623 & 6.3246 \\ 4.4721 & 13.4163 \\ 2.2361 & 2.2361 \\ 1.8257 & 9.1285 \\ 2.58199 & 23.2379 \end{bmatrix}, \quad y^* = W^{1/2} y = \begin{bmatrix} 12.649 \\ 31.3047 \\ 6.7083 \\ 16.4313 \\ 43.033 \end{bmatrix}$$

Step 4: Compute GLS coefficients

$$\hat{\beta}_{GLS} = (X^{*\top} X^*)^{-1} X^{*\top} y^*$$

Compute components:

$$X^{*\top} X^* \approx \begin{bmatrix} 45 & 161.667 \\ 161.667 & 848.33 \end{bmatrix}, \quad X^{*\top} y^* \approx \begin{bmatrix} 336.1 \\ 1665 \end{bmatrix}$$

$$(X^{*\top} X^*)^{-1} = \frac{1}{12037.25} \begin{bmatrix} 848.33 & -161.667 \\ -161.667 & 45 \end{bmatrix}$$

Multiply to get:

$$\hat{\beta}_0 \approx 0.133, \quad \hat{\beta}_1 \approx 1.716$$

Step 5: GLS regression equation

$$\boxed{y \approx 0.133 + 1.716x}$$

2.3 Solution Q3.

Step 1: WLS formula

For regression through the origin with known diagonal variance matrix $\Sigma = \text{diag}(\sigma_1^2, \sigma_2^2)$, the WLS estimator is:

$$\hat{\beta}_{WLS} = \frac{\sum_{i=1}^n w_i x_i y_i}{\sum_{i=1}^n w_i x_i^2}, \quad w_i = \frac{1}{\text{Var}(e_i)}$$

Here,

$$w_1 = \frac{1}{\sigma^2}, \quad w_2 = \frac{1}{2\sigma^2}$$

Step 2: Compute numerator and denominator

Numerator:

$$\sum_{i=1}^2 w_i x_i y_i = w_1 x_1 y_1 + w_2 x_2 y_2 = \frac{1}{\sigma^2}(y_1) + \frac{1}{2\sigma^2}(-y_2) = \frac{y_1 - \frac{1}{2}y_2}{\sigma^2}$$

Denominator:

$$\sum_{i=1}^2 w_i x_i^2 = w_1 x_1^2 + w_2 x_2^2 = \frac{1}{\sigma^2} + \frac{1}{2\sigma^2} = \frac{1.5}{\sigma^2}$$

Step 3: WLS estimate of β

$$\hat{\beta}_{WLS} = \frac{\text{Numerator}}{\text{Denominator}} = \frac{\frac{y_1 - \frac{1}{2}y_2}{\sigma^2}}{\frac{1.5}{\sigma^2}} = \frac{y_1 - \frac{1}{2}y_2}{1.5} = \frac{2}{3} \left(y_1 - \frac{1}{2}y_2 \right) = \frac{2y_1 - y_2}{3}$$

Step 4: Variance of $\hat{\beta}_{WLS}$

For WLS,

$$\text{Var}(\hat{\beta}_{WLS}) = \frac{1}{\sum_{i=1}^n w_i x_i^2} = \frac{1}{1.5/\sigma^2} = \frac{2}{3}\sigma^2$$

Step 5: Final Answer

$$\boxed{\hat{\beta}_{WLS} = \frac{2y_1 - y_2}{3}, \quad \text{Var}(\hat{\beta}_{WLS}) = \frac{2}{3}\sigma^2}$$